Supplemental Material: What static transport data determine about a conserving phonon closure

Koussay Mansouri[1]

[1] *[**PENDING:** affiliation; author list and order to be agreed]*

(Dated: November 12, 2026)

[**PENDING:** content of every section below]

S1. EVENT DECOMPOSITION OF THE THREE-PHONON OPERATOR AND ENERGY-CONSERVING STOICHIOMETRY

S2. LIFETIMES AND DC RESPONSE: SHARPNESS OF THE BOUND $K_0 \geq K_{\mathrm{RTA}}/\varsigma_{\max}$ AND SELF-ENERGY LINEWIDTHS

S3. LIFETIME-CONSTRAINED SHARPNESS OF THE CONE BOUND: AN EXACT SEVEN-MODE EXAMPLE

S4. INFRARED CRITERION: LATTICE SUMS AND MESH LAWS

S5. EVENT GEOMETRIES: DEBYE-TYPE MESHES AND ALN

S6. FEASIBLE RANGES OF THE MEMORY MOMENT: OPTIMIZATION, CERTIFIED WITNESSES AND IMPLICIT-FUNCTION TRANSFER

S7. FRONT-FACE QUADRUPOLE: DERIVATION, NUMERICAL STABILITY AND INDEPENDENT CHECKS

S8. RANK THEOREM: PROOF AND COMPUTER-ALGEBRA VERIFICATION

S9. FDTR MODEL: TRANSDUCER, INTERFACES, BEAMS, NUISANCE SETS AND CONVERGENCE

S10. REPRODUCIBILITY: ENVIRONMENT, INPUTS, SCRIPTS AND CHECKSUMS

S11. FAILED APPROACHES THAT AFFECT THE INTERPRETATION